Uniwersytet SWPS

Task list no. 11 – In this lab, we will implement a Huffman tree for data compression and also work on Breadth-First Search (BFS) as a graph search algorithm.

Task 1: Implement a greedy algorithm to construct Huffman codes for data compression for the following symbols. Please print the minheap after each insert/delete, and also, print the code for each character after your tree is created.

Symbol= { 'f', 'e' ,'c', 'b', 'd', 'a', 'z', 'm' , 'n', 'p'}
Freq = { 5, 9, 12, 13, 16, 45, 1, 25, 30 , 17 }

Task 2. Write the Breadth-First Search (BFS) algorithm. Then, apply the algorithm to the following adjacency matrix. Print the vertices in the order of their discovery during the search. Also, for each discovered vertex, print its shortest-path distance from the source vertex and its predecessor (parent).

	S(0)	R(1)	U(2)	V(3)	T(4)	W(5)	Y(6)	X(7)	Z(8)
S(0)	0	1	1	1	0	0	0	0	0
R(1)	1	0	0	0	1	1	0	0	0
U(2)	1	0	0	0	1	0	1	0	0
V(3)	1	0	0	0	0	1	1	0	0
T(4)	0	1	1	0	0	0	0	0	0
W(5)	0	1	0	1	0	0	0	1	1
Y(6)	0	0	1	1	0	0	0	1	0
X(7)	0	0	0	0	0	1	1	0	1
Z(8)	0	0	0	0	0	1	0	1	0

Note. ALL PROGRAMS SHOULD COMPILE WITHOUT ERROR AND WORK (i.e. DO EXACTLY WHAT IS DESCRIBED IN THE TASK)!!!

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